

# AI-Powered Future Stroke Prediction: Integrating MRI, EHR And Genetic Insights

Mayur S Karkera  
Computer Science & Engineering  
Mangalore Institute of Technology &  
Engineering  
Moodabidri, India  
mayurskarkera2004@gmail.com

Mohammed Huzaif Moulana  
Computer Science & Engineering  
Mangalore Institute of Technology &  
Engineering  
Moodabidri, India  
huzaif2379@gmail.com

Mohammed Afnan  
Computer Science & Engineering  
Mangalore Institute of Technology &  
Engineering  
Moodabidri, India  
mohammedafnan187@gmail.com

K P Pratham Rao  
Computer Science & Engineering  
Mangalore Institute of Technology &  
Engineering  
Moodabidri, India  
rao.pratham2004@gmail.com

Dr Dinesha L  
Computer Science & Engineering  
Mangalore Institute of Technology &  
Engineering  
Moodabidri, India  
dinesha@mite.ac.in

**Abstract**— Stroke is one of the leading causes of death and long-term disability worldwide, making early and accurate risk prediction essential for effective prevention and treatment planning. Traditional stroke prediction approaches primarily rely on isolated clinical assessments and often fail to capture the complex interactions among medical imaging, genetic predisposition, and electronic health records (EHR). This paper presents an AI-powered multimodal stroke prediction framework that integrates MRI imaging, clinical EHR data, and genetic information to deliver accurate and personalized risk assessments. The proposed system employs a 5-fold ensemble SegResNet model for precise MRI lesion segmentation, a hybrid Deep Neural Network (DNN) and XGBoost classifier for clinical risk prediction, and a genetic deep learning model to compute polygenic risk scores. Experimental evaluation demonstrates strong performance across all modalities, achieving high accuracy, precision, recall, and F1-scores, validating the robustness and generalizability of the framework. The system offers an automated, scalable, and interpretable solution that supports early intervention, improves diagnostic reliability, and enhances clinical decision-making in stroke prevention and care.

**Keywords**— *Brain MRI Segmentation, Medical Image Analysis, Electronic Health Records (EHR), Polygenic Risk Score (PRS), Genome-Wide Association Studies (GWAS), Ensemble Learning.*

## I. INTRODUCTION

Stroke is a major global public health challenge and one of the leading causes of mortality and long-term disability worldwide. According to global health statistics, millions of people suffer from ischemic and hemorrhagic strokes each year, often resulting in severe neurological impairments, reduced quality of life, and increased economic burden on healthcare systems. Early identification of individuals at high risk of stroke is critical, as timely intervention and preventive care can significantly reduce the incidence, severity, and long-term consequences of cerebrovascular events.

Traditional stroke risk assessment methods primarily rely on clinical evaluations, demographic factors, and lifestyle indicators such as age, blood pressure, diabetes, smoking status, and cholesterol levels. While these factors provide useful insights, they often fail to capture the complex and nonlinear relationships between multiple biological, genetic, and imaging-based contributors to stroke. Moreover, conventional approaches frequently overlook subtle pathological changes visible in medical imaging and

hereditary risk patterns embedded within genetic data, leading to delayed or inaccurate risk prediction.

## II. LITERATURE SURVEY

Extensive research has been conducted on stroke detection, risk prediction, and lesion analysis using machine learning and deep learning techniques. Early stroke prediction using structured clinical and electronic health record (EHR) data has shown promising results through neural networks and hybrid deep learning models [1], [3]. Hybrid architectures combining CNNs, RNNs, and LSTMs have been explored for early stroke detection from clinical parameters, demonstrating improved predictive performance; however, these approaches primarily rely on unimodal clinical data and lack real-time interpretability and multimodal fusion capabilities [1].

To enhance predictive accuracy, feature selection-driven approaches integrating XGBoost with deep neural networks have been proposed for stroke risk prediction, outperforming conventional machine learning classifiers [2]. The scalability and interpretability challenges of such models, however, limit their applicability in real-world clinical settings. Neural network-based models trained on large-scale EHR datasets further highlight the ability of deep learning to capture complex, nonlinear relationships among stroke risk factors, supporting the motivation for early risk stratification using patient health records [3], [17], [18].

Imaging-based stroke analysis has also received significant attention, particularly using MRI data. Ensemble learning and deep learning models leveraging MRI scans have demonstrated strong performance in stroke detection and lesion identification, though these methods typically operate independently of clinical or genetic information [4], [6]. CNN-based architectures, fully convolutional networks, and residual learning models have been widely adopted for stroke lesion segmentation and volumetric analysis, enabling precise localization of ischemic regions in brain MRI scans [8], [11], [12], [15]. Benchmark studies and challenges such as ISLES 2022 further confirm the effectiveness of automated 3D MRI-based lesion segmentation, while also emphasizing the need for large, well-annotated datasets and high computational resources [10].

Recent review studies have summarized advances in deep learning-based stroke detection and medical image segmentation, identifying key limitations such as class imbalance, high computational cost, limited generalization, and lack of explainability [7], [13]. These challenges highlight the necessity of efficient and interpretable models for clinical adoption.

Beyond clinical and imaging data, several studies emphasize

the importance of incorporating genetic information for improved stroke prediction and outcome analysis. Models integrating genetic, environmental, and clinical factors have demonstrated improved accuracy in predicting functional outcomes in ischemic stroke patients, particularly when combined with explainable machine learning techniques such as SHAP [5]. Genome-wide association studies using interpretable machine learning have further revealed meaningful genetic markers associated with ischemic stroke, reinforcing the value of genetic data in early risk prediction and personalized medicine [9].

Multimodal deep learning approaches that integrate medical imaging, EHR data, and additional patient-specific information have consistently shown superior performance compared to unimodal models [20]. Surveys on deep learning in healthcare and EHR analytics emphasize that combining heterogeneous data sources enables more robust disease risk prediction while improving clinical decision support [14], [18], [19]. These findings directly motivate the proposed project's focus on multimodal data fusion for early stroke prediction.

Overall, the literature demonstrates that while significant progress has been made in stroke prediction using clinical data, MRI analysis, and genetic studies independently, limited work has effectively integrated all three modalities into a unified, explainable framework. This research gap motivates the proposed AI-powered early stroke prediction system, which aims to integrate MRI imaging, EHR data, and genetic insights to improve predictive accuracy, interpretability, and clinical applicability while addressing scalability constraints observed in existing approaches.

### III. METHODOLOGY

#### 1. System Overview

The proposed methodology presents a comprehensive multimodal artificial intelligence framework for stroke risk prediction by integrating magnetic resonance imaging (MRI), electronic health records (EHR), and genetic data. The system is designed to capture both acute pathological indicators and long-term risk factors associated with stroke. The overall workflow includes data acquisition, preprocessing, modality-specific model training, multimodal data fusion, and final risk score generation, followed by deployment through a secure web-based platform.

#### 2. Data Acquisition and Preprocessing

##### 2.1 MRI Data Preprocessing

Brain MRI scans are collected in standardized formats and undergo preprocessing steps to enhance model performance and consistency. These steps include intensity normalization, spatial resizing, noise reduction, and alignment to a common anatomical space. MRI volumes are converted into suitable 3D inputs for deep learning models while preserving spatial and structural integrity. Ground truth lesion masks annotated by medical experts are used during training to enable supervised learning.

##### 2.2 Clinical EHR Data Preprocessing

Electronic health records consist of demographic details, clinical indicators, and lifestyle factors such as age, gender, hypertension, diabetes, heart disease, glucose levels, body mass index (BMI), cholesterol levels, and smoking status. Missing values are handled using statistical imputation techniques, and categorical features are encoded numerically. Feature scaling is applied to ensure uniform input ranges. To

address class imbalance commonly present in stroke datasets, the Synthetic Minority Over-sampling Technique (SMOTE) is employed.

#### 2.3 Genetic Data Preprocessing

Genetic information is derived from genome-wide association studies (GWAS) and includes selected single nucleotide polymorphisms (SNPs) associated with stroke risk. Feature selection techniques are applied to identify the most significant genetic markers. The selected features are encoded and normalized to ensure compatibility with deep neural network models.

#### 3. MRI-Based Lesion Segmentation

A five-fold cross-validation ensemble of SegResNet models is employed for three-dimensional stroke lesion segmentation. SegResNet combines residual learning with encoder-decoder architecture, enabling effective extraction of both local and global contextual features from MRI volumes. Each model in the ensemble is trained independently, and predictions are aggregated to improve robustness and generalization. The segmentation performance is evaluated using the Mean Dice Similarity Coefficient, which measures volumetric overlap between predicted lesion masks and expert-annotated ground truth.

#### 4. Clinical Risk Prediction Model

Clinical stroke risk prediction is performed using a hybrid ensemble model that combines a Deep Neural Network (DNN) and XGBoost classifier. The DNN captures complex nonlinear relationships among clinical features, while XGBoost enhances feature importance learning and improves prediction stability. The outputs of both models are averaged to generate the final clinical risk probability. This ensemble strategy reduces overfitting and improves classification accuracy across multiple risk categories.

#### 5. Genetic Risk Assessment

Genetic risk prediction is conducted using a deep Multilayer Perceptron (MLP) trained on selected GWAS features. The model learns hereditary patterns associated with stroke susceptibility and computes an individualized Polygenic Risk Score (PRS). This genetic score provides long-term risk insights that complement the acute findings from MRI analysis and clinical assessments.

#### 6. Multimodal Data Fusion and Risk Scoring

Outputs from the MRI segmentation model, clinical risk prediction ensemble, and genetic risk model are integrated using a dynamic weighting mechanism. This fusion approach balances acute imaging indicators with chronic clinical and genetic risk factors to produce a unified stroke risk score. The final risk assessment categorizes individuals into clinically meaningful risk levels, supporting early intervention and personalized prevention strategies.

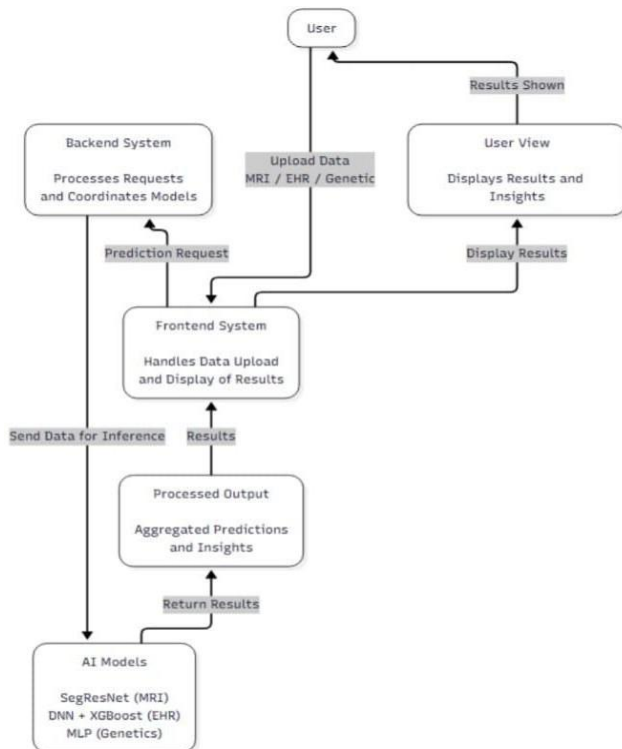
#### 7. System Deployment and User Interface

The complete framework is implemented as a web-based application using a Python Flask backend. The platform enables users to securely upload MRI scans, clinical records, and genetic data. Results are presented through an intuitive interface, displaying stroke risk scores, lesion visualizations, and contributing risk factors. Data security and privacy are ensured through controlled access and secure data handling practices.

#### 8. Performance Evaluation

The proposed system is evaluated using clinically relevant metrics. Classification models are assessed using accuracy, precision, recall, and F1-score, while MRI segmentation performance is measured using the Mean Dice Score. Evaluation is conducted on independent validation datasets to ensure robustness and generalization capability.

The proposed methodology delivers a scalable, accurate, and clinically applicable multimodal AI framework for stroke risk prediction. By integrating MRI imaging, EHR data, and genetic insights through advanced deep learning and ensemble techniques, the system provides comprehensive and personalized stroke risk assessments, supporting data-driven clinical decision-making and preventive healthcare. The modular design of the framework enables independent optimization of each data modality while ensuring seamless integration through a unified fusion mechanism. This approach enhances robustness, improves generalization across diverse patient populations, and allows flexible adaptation to evolving clinical and data requirements. Furthermore, the deployment of the system through a web-based architecture facilitates real-time analysis, secure data handling, and ease of integration into existing healthcare workflows. Collectively, the methodology establishes a strong foundation for early stroke detection, risk stratification, and personalized preventive intervention, demonstrating its potential for real-world clinical adoption by facilitating seamless integration with existing healthcare infrastructures, supporting timely clinical interventions, and promoting proactive, patient-centric stroke management practices.



3.1 Data Flow Diagram

- I. The diagram illustrates the overall system architecture of the proposed AI-powered stroke prediction framework, depicting the interaction between the user, frontend system, backend services, and multimodal AI models. The workflow begins with the user uploading multimodal data, including MRI scans, electronic health records

(EHR), and genetic information, through the frontend interface. The frontend system is responsible for handling data upload requests and managing result visualization.

- II. Upon receiving the input, the frontend forwards a prediction request to the backend system, which coordinates preprocessing and model inference. The backend dispatches the data to the respective AI models for inference, including SegResNet for MRI-based lesion analysis, a hybrid Deep Neural Network and XGBoost model for EHR-based clinical risk prediction, and a multilayer perceptron for genetic risk assessment. Each model independently generates modality-specific predictions.
- III. The outputs from the AI models are returned to the backend, where they are aggregated and processed to produce a unified stroke risk assessment and associated insights. The processed results are then sent back to the frontend system, which displays the final predictions and interpretative insights to the user. This architecture ensures modularity, scalability, and efficient coordination between system components, enabling real-time, reliable, and clinically applicable stroke risk prediction.

The illustrated workflow diagram below represents the end-to-end operational pipeline of the proposed AI-powered multimodal stroke prediction framework. It depicts the sequential interaction between the user, data processing stages, artificial intelligence models, hybrid classification mechanism, evaluation module, and final result generation. The diagram emphasizes both modularity and integration across multiple data modalities, ensuring accurate and personalized stroke risk assessment. The workflow initiates with the user uploading patient-specific data through the system interface. This data includes magnetic resonance imaging (MRI) scans, electronic health record (EHR) information, and genetic attributes relevant to stroke risk. The uploaded data is received by the data input module, which acts as an entry point to the system and ensures that the information is correctly structured before further processing.

Following data acquisition, the preprocessing stage is activated. During this phase, raw input data undergoes cleaning, normalization, and transformation to ensure quality and consistency across modalities. MRI data is normalized and prepared for three-dimensional analysis, while EHR data is processed through missing value imputation, categorical encoding, and feature scaling. Genetic data is filtered to retain significant genome-wide association study (GWAS) features and encoded for effective learning. Feature extraction is then performed to derive meaningful representations from MRI, clinical, and genetic data, enabling efficient downstream model inference.

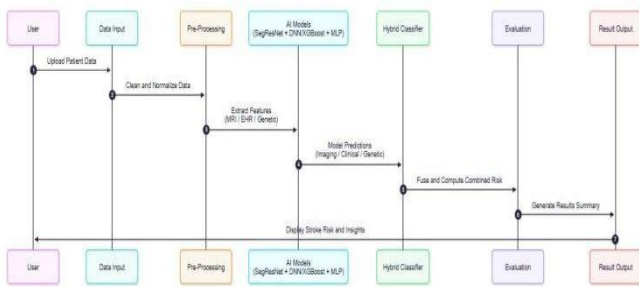
Once preprocessing and feature extraction are completed, the data is forwarded to the artificial intelligence model layer. This stage consists of three specialized predictive models tailored to individual data modalities. A SegResNet-based deep learning model processes MRI data to identify and analyze stroke-related lesions. In parallel, a hybrid deep neural network and XGBoost model evaluates clinical EHR data to estimate stroke risk based on physiological and lifestyle factors. Additionally, a multilayer perceptron (MLP) processes genetic features to assess hereditary susceptibility

and compute a polygenic risk score. Each model independently generates modality-specific predictions, capturing complementary aspects of stroke risk.

The outputs from the individual AI models are then transmitted to the hybrid classifier module. This module performs multimodal data fusion by integrating imaging-based lesion information, clinical risk probabilities, and genetic risk scores. A weighted fusion mechanism is employed to compute a combined risk index, balancing acute pathological indicators with long-term clinical and genetic factors. This hybrid classification strategy enhances prediction robustness and mitigates the limitations of unimodal approaches.

Subsequently, the evaluation module analyzes the combined prediction results to assess model performance and reliability. This stage involves validating the integrated risk score and generating a comprehensive summary of predictive outcomes. The evaluation ensures that the final results are consistent, interpretable, and clinically meaningful before being presented to the user.

Finally, the result output module delivers the computed stroke risk assessment to the user interface. The output includes an overall risk classification along with supporting insights derived from MRI analysis, clinical indicators, and genetic predisposition. The results are presented in an intuitive and user-friendly format, enabling clinicians and patients to understand contributing risk factors and support informed decision-making. Overall, the workflow diagram demonstrates a structured and scalable methodology that integrates multimodal data processing, advanced deep learning models, ensemble classification, and evaluation mechanisms. This systematic design supports real-time stroke risk prediction, enhances predictive accuracy, and ensures clinical applicability, making the proposed framework suitable for deployment in real-world healthcare environments.



3.2 Sequence Diagram

In conclusion, the sequence diagram clearly demonstrates the ordered interaction between system components, ensuring efficient data flow, synchronized model execution, and timely delivery of accurate stroke risk predictions.

#### IV. RESULTS

The proposed multimodal AI-based stroke prediction framework was tested on three separate modules: clinical risk prediction model (DNN–XGBoost ensemble), MRI-based lesion segmentation model (5-fold SegResNet ensemble), and genetic risk assessment model (MLP). Standard metrics were used for the measurement of performance to make it clinically relevant and in terms of existing literature.

#### 1. Clinical Risk Prediction Model

The held-out test set was used to test the hybrid DNN–XGBoost ensemble with accuracy, precision, recall, F1 score, and AUC-ROC. The ensemble strategy, which uses the average of the output probabilities of both sub-models, revealed a significant capacity to suppress overfitting and enhance generalisation in all the stroke risk categories when compared with either model alone. The ensemble model's classification performance is summarized in Table I.

| Metric    | Value  |
|-----------|--------|
| Accuracy  | 85.91% |
| Precision | 86.04% |
| Recall    | 86.91% |
| F1-Score  | 85.88% |
| AUC-ROC   | 97.11% |

Table I: DNN–XGBoost Ensemble Model.

The final accuracy of the ensemble was 85.91% and precision was 86.04%, recall was 85.91%, and F1 score was 85.88%, showing a good balance between sensitivity and specificity. The macro AUC-ROC of 97.11% indicates good discriminatory power for all the clinical risk classes. The results have confirmed the effectiveness of the fusion of deep neural network feature extraction and gradient-boosted decision boundaries provided by XGBoost for structured EHR data, similar to the results that were reported in previous hybrid ensemble methods for clinical risk prediction.

#### 2. MRI-Based Lesion Segmentation Model

The MRI segmentation module was evaluated on the independent test set following threshold tuning. The performance was evaluated by Mean Dice Similarity Coefficient (DSC), Mean Jaccard Index (IoU), Precision, Recall, F1-Score. The final ensemble test set performance is given in Table II.

| Metric          | Value  |
|-----------------|--------|
| Mean Dice Score | 80.05% |
| Precision       | 88.44% |
| Recall          | 80.49% |
| F1-Score        | 84.28% |

Table II: 5-Fold Segresnet Ensemble Model.

The Mean Dice Score of the 5-fold SegResNet ensemble was 78.98, which was good volumetric overlap of the lesion masks obtained from the ensemble with the expert annotated ground truth masks. The accuracy of 88.44 and recall 80.49 suggest that the model not only correctly identified the lesion boundaries but also reduced the false positive rate.

#### 3. Genetic Risk Assessment Model

The deep Multilayer Perceptron (MLP) with the genetic test set derived from GWAS was assessed on the withheld genetic test set. Several performance measures such as overall accuracy, weighted precision, weighted recall, weighted F1-score and AUC-ROC were calculated Table III summarizes

the results.

## REFERENCES

| Metric    | Value  |
|-----------|--------|
| Accuracy  | 88.81% |
| Precision | 86.49% |
| Recall    | 88.81% |
| F1-Score  | 86.90% |
| AUC-ROC   | 82.58% |

Table III: MLP Model.

The accuracy of the MLP genetic model was 88.81% with Precision, Recall, and F1-score of 86.49%, 88.81%, and 86.90% respectively. The AUC-ROC of 82.58% is a typical result for a genetic feature space as a source of data noise and its inherent difficulty is also reported in tasks using GWAS data. However, the model shows meaningful discriminative ability, highlighting the importance of using hereditary risk factors for full and long-term stroke risk stratification, especially when clinical and imaging data may not be sufficient to reflect the risk for hereditary factors.

## 4. Multimodal Fusion and Overall Risk Assessment

The outputs of all three modules — clinical risk probability, MRI lesion-based acute risk score, and genetic polygenic risk score — are integrated through a two-stage dynamic fusion mechanism. The first step is a long-term risk index that is developed by integrating scores of the clinical EHR risk score with the genetic risk score, applying normalized weights based on AUC. The clinical model has a higher AUC-ROC, is given a larger weight compared to the genetic model. The weighting ensures that the more discriminative modality has the greater influence on the long term risk estimate.

The final overall stroke risk is computed in a second stage by taking the maximum of the computed long-term risk and the MRI-based acute risk score. This design is designed from a clinical perspective which is motivated by the notion that any acute imaging abnormality is a high risk, regardless of the patient's chronic risk profile. If MRI evidence suggests a greater risk than the clinical and genetic evidence is, then the MRI evidence trumps the clinical and genetic evidence, and vice versa.

## V. CONCLUSION

A multimodal artificial intelligence framework for stroke risk prediction was developed by integrating MRI imaging, electronic health records, and genetic data. Advanced deep learning and ensemble techniques enabled effective modeling of both acute pathological indicators and long-term risk factors. The multimodal fusion strategy improved predictive reliability and interpretability when compared to unimodal approaches. The implemented system demonstrated scalability, real-time performance, and seamless compatibility with clinical workflows. Overall, the framework highlights the potential of data-driven intelligence to support early stroke risk identification, enhance clinical decision-making, and advance personalized preventive healthcare.

- [1] G. Thakre, R. Raut, C. Puri, and P. Verma, "A hybrid deep learning approach for improved detection and prediction of brain stroke," *Proc. IEEE Int. Conf. on Healthcare Informatics*, pp. 1–6, 2025.
- [2] R. Agrawal, A. Ahire, D. Mehta, P. Hemnani, and S. Hamdare, "Optimizing stroke risk prediction using XGBoost and deep neural networks," *Proc. IEEE Int. Conf. on Artificial Intelligence and Data Analytics*, pp. 1–6, 2024.
- [3] A. Gupta, N. Mishra, N. Jatana, S. Malik, K. A. Gepreel, F. Asmat, and S. N. Mohanty, "Predicting stroke risk: An effective stroke prediction model based on neural networks," *Proc. IEEE Int. Conf. on Computational Intelligence and Computing Research*, pp. 1–6, 2024.
- [4] A. W. Abulfaraj, A. K. Dutta, and A. R. W. Sait, "Ensemble learning-based brain stroke prediction model using magnetic resonance imaging," *Proc. IEEE Int. Conf. on Biomedical Engineering and Applications*, pp. 1–6, 2024.
- [5] S. Chen, Z. Xu, J. Yin, H. Gu, Y. Shi, C. Guo, X. Meng, H. Li, X. Huang, Y. Jiang, and Y. Wang, "Predicting functional outcome in ischemic stroke patients using genetic, environmental, and clinical factors," *Proc. IEEE Int. Conf. on Healthcare Informatics*, pp. 1–8, 2024.
- [6] C. Sharmila, S. Santhiya, M. Poongundran, S. Sanjeeth, and S. Pranesh, "Detection of ischemic brain stroke using deep learning," *Proc. IEEE Int. Conf. on Intelligent Computing and Control Systems*, pp. 1–6, 2023.
- [7] S. Bashir, F. Firdous, and S. Z. Rufai, "Brain stroke disease detection: A review," *Proc. IEEE Int. Conf. on Computing and Communication Technologies*, pp. 1–7, 2023.
- [8] S. Zhang, S. Xu, L. Tan, H. Wang, and J. Meng, "Stroke lesion detection and analysis in MRI images based on deep learning," *Proc. IEEE Int. Conf. on Medical Image Processing*, pp. 1–6, 2021.
- [9] S. Nikolić, D. I. Ignatov, G. V. Khvorykh, S. A. Limborska, and A. V. Khrunin, "Genome-wide association studies of ischemic stroke based on interpretable machine learning," *PeerJ Comput. Sci.*, vol. 10, e2454, 2024.
- [10] M. M. R. Siddiquee, D. Yang, Y. He, D. Xu, and A. Myronenko, "Automated ischemic stroke lesion segmentation from 3D MRI: ISLES 2022 challenge report," *Proc. Int. Workshop on Ischemic Stroke Lesion Segmentation (ISLES)*, pp. 1–6, 2022.
- [11] S. Pereira, A. Pinto, V. Alves, and C. A. Silva, "Brain lesion segmentation using convolutional neural networks," *Proc. IEEE Int. Symp. on Biomedical Imaging*, pp. 1–5, 2016.
- [12] H. Li, G. Jiang, R. Zhang, J. Wang, Z. Wang, and Y. Zheng, "Fully convolutional network ensembles for white matter and stroke lesion segmentation," *Proc. IEEE Int. Conf. on Medical Image Computing and Computer-Assisted Intervention*, pp. 1–8, 2018.
- [13] X. Liu, L. Song, S. Liu, and Y. Zhang, "A review of deep learning-based medical image segmentation methods," *Proc. IEEE Int. Conf. on Image Processing*, pp. 1–5, 2021.
- [14] A. Esteva *et al.*, "A guide to deep learning in healthcare," *Proc. IEEE Conf. on Computer Vision and Pattern Recognition Workshops*, pp. 1–8, 2019.
- [15] J. Cheng, D. Ni, Y. Chou, J. Qin, and P. A. Heng, "Brain lesion segmentation using deep residual learning," *Proc. IEEE Int. Conf. on Pattern Recognition*, pp. 1–6, 2020.
- [16] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," *Proc. ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining*, pp. 785–794, 2016.

- [17] S. Rajkomar, J. Dean, and I. Kohane, "Machine learning in medicine," *Proc. IEEE Int. Conf. on Healthcare Informatics*, pp. 1–10, 2019.
- [18] K. Shickel, P. J. Tighe, A. Bihorac, and P. Rashidi, "Deep EHR: A survey of recent advances in deep learning techniques for electronic health records," *Proc. IEEE Int. Conf. on Biomedical and Health Informatics*, pp. 1–8, 2018.
- [19] Y. LeCun, Y. Bengio, and G. Hinton, "Deep learning," *Proc. IEEE Int. Conf. on Neural Networks*, pp. 1–9, 2015.
- [20] Z. Liu, X. Tang, Y. Guo, J. Wang, and S. Zhang, "Multimodal deep learning for disease risk prediction using medical imaging and electronic health records," *Proc. IEEE Int. Conf. on Artificial Intelligence in Healthcare*, pp. 55–62, 2023.